UNIT 1 · LESSON 3

Light Waves

What light is, the electromagnetic spectrum, and why things have color.

In this lesson you will learn...

- Why light doesn't need a medium
- The electromagnetic spectrum
- The speed of light, and energy vs. frequency
- Why objects look the color they do

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Words to know

Light
Energy that travels as a wave and that our eyes can see.

EM wave
A wave made of electric and magnetic fields. It needs no medium.

EM spectrum
All EM waves in order, from radio waves to gamma rays.

Visible light
The small part of the spectrum our eyes can see: the rainbow colors.

Reflect / absorb
Bounce light back / soak light up (it turns into heat).

Brightness
For light, a bigger amplitude means brighter light.

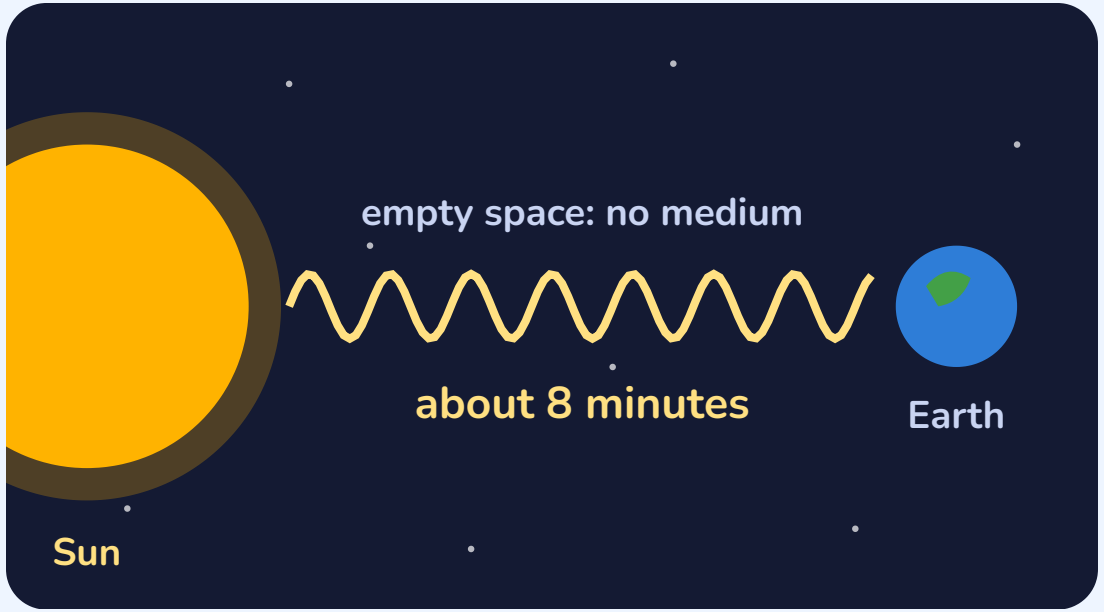
Think about it

On a sunny day, you wear a black T-shirt and your friend wears a white one.

Who feels hotter, and why?

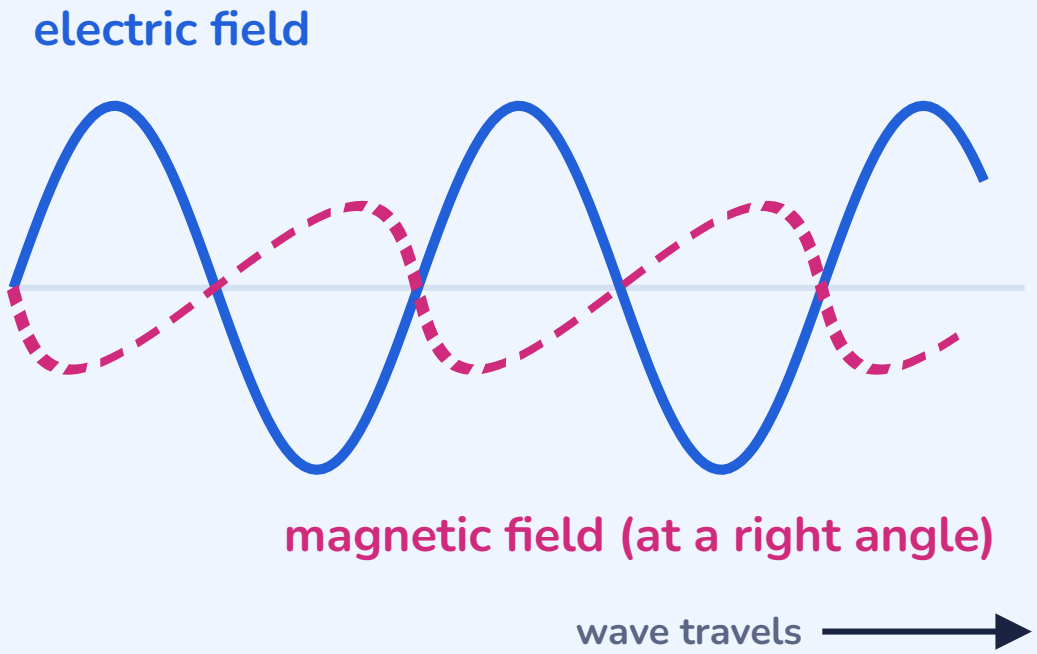
Light doesn't need a medium

- Light is an **electromagnetic (EM) wave**.
- It can travel through completely empty space.
- Sunlight crosses about 150 million km of space to reach us. The trip takes about 8 minutes.
- Sound could never do that: it needs a medium.



What is an EM wave?

- When a charged particle vibrates, it shakes the electric and magnetic fields around it.
- The two fields vibrate at **right angles** to each other.
- Together they carry energy away, even through a vacuum.
- Light is a **transverse** wave.



The electromagnetic spectrum



Radio | Micro-waves | Infrared | Visible | Ultra-violet | X-rays | Gamma



← long wavelength · low frequency · less energy

short wavelength · high frequency · more energy →

💡 Visible light is only a tiny slice of the spectrum. Every other EM wave is invisible to us.

The speed of light

- In empty space, **every** EM wave travels at the same speed: **300,000,000 m/s**.
- That's fast enough to go around Earth about 7½ times in one second!
- Light slows down a little in materials like water and glass.

c = 300,000,000 m/s

💡 Radio, light and X-rays all travel at this same speed in space. They differ in wavelength and frequency.

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Energy and frequency

- **Higher frequency** (shorter wavelength) EM waves carry **more energy**.
- High-energy waves can harm living things. Too much UV gives you sunburn, and X-ray technicians stand behind shields.
- Radio waves have low frequency and little energy, so they're safe all around us.

Radio
low frequency · low energy · safe

Ultraviolet
higher frequency · sunburn

X-rays
very high frequency · use with care

Colors of visible light

- Each color has its own wavelength and frequency.
- Remember the order with **ROY G. BIV**: red, orange, yellow, green, blue, indigo, violet.
- **Red**: longest wavelength, lowest frequency.
- **Violet**: shortest wavelength, highest frequency.

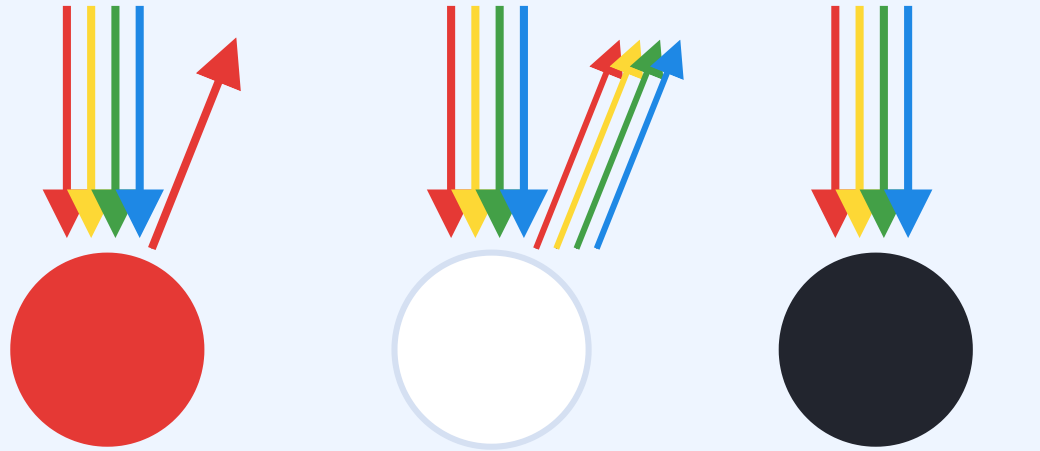


long λ
low f

short λ
high f

Why do things have color?

- When light hits an object, some colors are **reflected** and the rest are **absorbed**.
- You see the colors that bounce into your eyes.
- White reflects all colors. Black absorbs almost all of them.
- Absorbed light turns into heat. That's why the black T-shirt feels hotter!



red apple
reflects red

white shirt
reflects all colors

black shirt
absorbs all → hot

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The light source matters

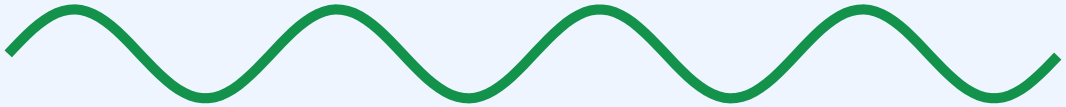
- An object can only reflect colors that are in the light hitting it.
- Under a red light, a blue shirt has no blue light to reflect, so it looks dark.
- That's why a street looks different at noon, at sunset and under street lamps.

💡 Try it: look at a colorful object under a phone flashlight, then under a colored bulb or screen. What changes?

Amplitude = brightness

- Light is a transverse wave with crests and troughs.
- Just like louder sound has bigger amplitude, **brighter light has bigger amplitude**.
- A candle and a stadium light both make light, but the stadium light's waves have much bigger amplitude: more energy, more brightness.

Candle: small amplitude



Stadium light: big amplitude



Check yourself

1. Why can we see the Sun but not hear it?
2. Put in order from lowest to highest energy: X-rays, radio waves, visible light.
3. A leaf looks green. Which color of light does it reflect? What happens to the others?
4. Which has the longer wavelength: red light or violet light?
5. Which doghouse stays cooler in summer, a white one or a black one? Explain.

Answers

1. Light is an EM wave and can cross empty space. Sound is mechanical and needs a medium.
2. Radio waves → visible light → X-rays.
3. It reflects green. The other colors are absorbed: the leaf uses some of that energy to make food, and the rest becomes heat.
4. Red light.
5. White. It reflects most light, while black absorbs it and turns it into heat.